Examiners' Report

Summer 2026

Pearson Edexcel GCE

In Further Mathematics (9FM0)

Paper 3A Further Pure Mathematics 1

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Summer 2026
Publication Code 9FM0_3A_2606_ER

General Comments

The paper provided accessible entry points for learners, with many responses demonstrating sound skills and effective application of the required methods. Routine tasks were generally completed well, and the paper also provided opportunities for careful thought, precision and rigour to be rewarded.

Some earlier questions proved more demanding, while some later questions were more accessible. Overall, the paper allowed learners to demonstrate a range of skills and provided effective discrimination across grades.

Presentation was generally clear, although some responses would have been improved by setting out working more carefully.

Individual Question Reports

Question 1

This opening question provided accessible entry points for learners. Most responses showed a clear understanding of the methods required, although some approaches were less direct than the intended route. In some responses, learners attempted to solve directly for x or y and then used these values to verify the equation, rather than making use of the structure indicated in part (a).

Part (a)

Part (a) was generally well answered, with many responses including detailed and complete working. Some learners first found a Cartesian equation for the hyperbola, whereas responses with simple substitution of the parameter for x and y, in terms of t, in the given equation were more successful. Accuracy was generally secure in the alternative where more extensive work was required, or in the variation of part (a) where an equation in just x or y was first found before substitution into this equation for the relevant parameter. A small number of responses showed difficulty when rearranging equations with variable terms in the denominators, but many learners handled this part efficiently.

Part (b)

Part (b) was also very well answered. Most responses found the coordinates of P and Q accurately, with working that was usually well structured and showed the steps leading to the answer. Where the final mark was not awarded, this was when finding one coordinate incorrectly (e.g. getting -6 instead of $-\frac{1}{6}$) or by mislabelling the two coordinate pairs.

Question 2

Overall, part (a) was answered well by most learners, while part (b) was more demanding. This gave an early opportunity for discrimination in the paper. In part (b), some responses indicated uncertainty about the meaning of the x-y plane, for example by not recognising that points in this plane have a zero z coordinate.

Part (a)

In part (a), most learners recognised $\mathbf{a}$ represented the position vector of a point on the line, and that $\mathbf{b}$ and $\mathbf{c}$ represented direction vectors. For the point on the line the most common vector used was the $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ from the given line equation, although $\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ was also frequently seen. Other valid points on the line were occasionally used. In some responses, however, the given information was not transferred directly to the problem, and a point on the line was used with a direction vector to form a third line rather than to define the plane.

The most common errors in this part were sign errors or other slips when subtracting vectors to find the second direction vector. Occasionally, responses used two points on the given line, resulting in two parallel direction vectors which do not define a plane. Very few responses omitted the $\mathbf{r} = \cdots$ form when giving the equation, although some did attempt the $\mathbf{r.n} = \mathbf{d}$ form of the equation of a plane instead; these responses could score at most two marks.

Part (b)

Part (b) was more demanding, with a variety of methods seen. Some responses made effective use of the method in the mark scheme (setting the z coordinate to 0, getting λ or μ in terms of the other and substituting back into the plane equation), although some learners used this method to link λ and μ, then chose values for each of these to find points on the line, which they then subtracted to find a direction vector. Where a correct approach was taken, errors sometimes occurred when rearranging the equation in λ and μ. In some responses, this led to equations with non-zero z components, indicating that the condition for points to lie in the x-y plane had not been applied correctly.

A common approach was to find a Cartesian equation for the plane, usually by using the cross product to find a normal vector and then determining d. This equation was then used to find two points before proceeding as above.

Where full marks were not achieved in this part, responses often showed uncertainty about the distinction between a direction vector and a normal vector, or about how the different forms of equations of lines and planes relate to each other. Some responses also showed uncertainty about the meaning of the x-y plane. For example, $\mathbf{r} = k\begin{pmatrix}1\\1\\0\end{pmatrix}$ was used as an equation for the x-y plane. These responses occasionally made progress by setting the z component to zero, although the subsequent method was not always completed correctly. In other responses, $x = 0$ or $y = 0$ was set instead of $z = 0$.

Question 3

This question provided good access for learners. Most responses showed sound command of the t-substitutions and were able to apply the substitution to solve the equation. The main points of difficulty occurred in part (b).

Part (a)

In part (a) most learners scored well. Learners were able to apply the correct t-formulae and the double-angle formulae for the $\frac{x}{2}$ and x terms, though some derived the identities, usually accurately, rather than quoting them. In general, the responses that used the double angle formula for $\sin 2A$ were more direct than those that used $\tan 2A$ as the latter required more algebraic manipulation to reach the same point. A small number of responses did not apply correct 13 -substitutions at all, and this usually because of confusion between the $\frac{x}{2}$ and $\frac{x}{4}$, and so applying incorrect identities to the relevant terms.

The main errors were less often in choosing the correct formulae, and more often in expanding brackets, handling signs, and cancelling terms correctly. Many responses showed good understanding of how to combine the resulting expressions into a single fraction shown, though. A small number of responses included slips in the final line and so did not reach the correct conclusion, but the algebra was usually handled well.

Part (b)

Part (b) was more demanding overall, although most responses were able to access at least the first mark. This part provided effective discrimination at the end of the question.

The first was to form a 3-term quadratic equation. Most responses proceeded by setting $H = 1.9$, bit errors were sometimes made when evaluating $1.9 - 1.65$ or when copying either the t or t^2 terms. Where this prevented a suitable 3 term quadratic equation from being formed, subsequent progress was more limited. Responses that formed a suitable equation could still access the next mark if the ensuing method was correct, although this was not always completed successfully.

Although many responses successfully solved the quadratic equation (or their resulting equation) in terms of t, some then misinterpreted the role of t, treating it as the time (the number of hours after midday) rather as the t -substitution. This indicated that, while the algebraic manipulation was often secure, interpreting the mathematical result in the context of the question was more demanding.

A common error in responses that went on to take the arctangent was not accounting correctly for $\frac{x}{4}$. This often led to responses such as 11 (or 22) minutes, even in otherwise well-structured responses.

Another common issue was that some responses correctly found the value ($x = 0.73$) but did not express this as the corresponding time. The context required this value to be converted to 12:44, whereas some responses gave a final answer of 44 minutes.

Question 4

This was the second vector question on the paper. Parts (a) and (b) provided good access for learners, while part (c) provided effective discrimination by requiring learners to think through and justify the solution carefully.

Part (a)

Part (a) was generally well answered, with most responses carrying out the cross product accurately. The most common error came from the **j**-component of the cross product, where sometimes the negative sign was omitted. Another common issue appeared to arise from unclear handwriting, with 9 sometime being read as q leading to a final entry of $pq - q$. Where this occurred, part (b) could usually access no more than the first mark. In some responses, the resulting expression in part (b) did not support the required conclusion, indicating that the earlier work in part (a) needed to be checked.

Part (b)

Again, part (b) was also well answered overall. Most responses used the magnitude of the vector product from part (a) correctly to find the area of the triangle. Very few responses omitted the factor of $\frac{1}{2}$ in front of the magnitude of the vector, so many were able to successfully set up the required equation and work through to the given result.

As this was a "Show that" question, responses needed to include sufficient work to justify the result. Where errors had been made in part (a), the pq terms did not cancel, so the second mark could not be awarded. In some responses, the working nevertheless continued to the printed result, indicating that the earlier algebra needed to be checked. Other responses omitted the pq terms altogether; as the cancellation had to be shown clearly, these responses could not be awarded the second mark. Learners should be reminded of the importance of showing sufficient working in this style of question.

Stronger responses showed the squared expression, expanded all terms, displayed the cancellation of the pq terms, and then arrived at the required equation.

Part (c)

Part (c) provided effective discrimination, as it required reasoning and a clear justification. The second mark, for identifying $p = 1$ and $q = 4$ as a solution, was the most frequently accessed, and was often the only mark awarded. It was uncommon for responses to miss this mark but gain the others. However, some responses gave only the $p = 0$ and $q = 5$ solutions or gave reasoning for the bounds on the values without identifying any valid case. Some responses also included negative cases.

Some responses overlooked the case $p = 0$ and $q = 5$, where the value 0 is a valid possibility because p and q are non-negative integers. In these responses, "non-negative" appeared to have been interpreted as positive, so this case was omitted. In some cases, this omission occurred even when negative values of p and q were also included as possible solutions.

The final mark was the least frequently accessed. Many responses identified possible cases but did not provide a complete explanation of why there are no further solutions. Some responses listed cases up to $p = 25$ without explaining why no larger values needed to be considered. To gain full credit, responses needed to justify that all possible cases had been considered.

The strongest responses used the restriction on non-negative integer values and explained why larger values of p would make q^2 negative or impossible. A few responses consider values for q first to derive a bound on p. The most effective approach was to establish a bound on the possible values and use this to justify that all cases have been considered. Whichever approach was taken, some kind of justification was required for full marks to be scored. Learners should be reminded to give clear reasoning when a question asks them to "justify" their answer.

Question 5

This was a routine question that provided good access for learners. Most responses showed a sound understanding of the process, but marks were sometimes limited by the interpretation of the units used in the model. Careful reading of the information given in the question was important. Many responses gained the middle marks by carrying out a correct process using one of the accepted starting conditions. Responses that identified the correct starting values often went on to gain full marks, with only occasional slips or answers not given in the required context preventing full credit.

The most common errors involved the starting value for V_0, with 225000 and 225 commonly used, instead of the correct value of 2.25. A few responses used degrees rather than radians in the calculations, which could potentially be awarded all except the final mark. Only a small number or responses used an incorrect value for h, usually 3. Other errors in the initial conditions were less common.

Most responses substituted their starting values correctly, carried out the required procedure accurately, and used calculators effectively. Many then gave the answer to the nearest thousand pounds, as required, including some indication of pounds in the final answer. A few responses did not complete this final contextual step.

Other errors were rare, but some calculations slips were observed, usually involving the derivative formula when substituting values, for example omitting to square root in the denominator. A small number of responses substituted incorrect values, e.g. the wrong t values in the derivatives, which limited access to the method marks. Where no substitution was shown but a stated value was given, benefit of doubt could sometimes be applied; however, learners should be reminded to show enough working to make their method clear rather than relying on implied method.

Question 6

This was another routine question that provided good access for learners. The main issue was the absence of set notation in many answers. Learners generally performed well in finding the critical values, particularly the two values other than -4, although a few responses omitted -4 or gave it as 4 instead.

Part (a)

A range of methods was seen for finding the critical values. Most responses multiplied through by $(x+4)^2$ and factorised. Using a common denominator was less common, and simply setting the two sides equal to find the critical values was seen less frequently. As this was not a "show that" question, a formal route was not required, and any valid method for finding the critical values was acceptable. Relatively few algebraic errors were seen when manipulating the equation. Where errors did occur, solving the resulting equation by a calculator method could still be acceptable, provided algebra had first been used to set up a suitable equation.

Once the critical values had been found, responses with three critical values were generally able to form a solution set of the correct shape. Responses that omitted the -4, could not access the later marks in part (a). Even where incorrect critical values were found, the method mark for the shape of the solution set could still be awarded if three suitable values were used, although a small number of responses selected the complementary region instead. For minor errors in the equation, the follow-through mark was often still available where the values were suitable to use.

The main issue in part (a) was the final form of the answer. Only a small number of responses gave the solution in set notation or as a clear union of intervals, so the final mark was not always awarded in otherwise correct solutions. Learners should be reminded of the importance of using set notation when the solution is required as a set.

Part (b)

In part (b), most responses were able to obtain the required inequality using the answer from part (a), including by follow-through where the solution to part (a) was in a suitable format. Many responses used graph sketches effectively, showing clearly that there was only one point of intersection and then using this to obtain the answer. Some responses restarted the process and repeated much of the work from part (a), rather than deducing the result from the previous answer. Other responses squared both sides of the modulus equation, which led to varying degrees of progress.

Effective use of sketches would have supported many responses in this part, as the answer to part (b) becomes clear once the sketch has been formed.

Question 7

This question required learners to connect a Taylor expansion with L'Hôpital's rule and to show that both methods gave the same value for the limit. Responses varied considerably. A large proportion of responses were successful, while others found it difficult to make sustained progress. Responses that were less successful in part (a) could often still access part (b).

Part (a)

Part (a) was generally accessible, with most responses differentiating their exponentials terms correctly up to and including the third derivative. Evaluations were typically correct, although a notable number of responses evaluated $f\left(\frac{1}{2}\right)$ to 1 or -1 instead of 0, while recognising $e^0 = 1$ in their derivatives. Although the expansion was provided, a small number of responses did not use it, and instead attempted to recall the expansion independently, with varying degrees of success. Responses that found the differentiation more demanding could still gain marks by applying the Taylor expansion correctly.

Part (b)

Part (b) was handled well by most responses, regardless of performance in part (a). Many responses were complete and well structured. Where differentiation errors had occurred in part (a), these were often repeated in part (b), but the first two marks could still be accessed if the denominator was differentiated successfully. A small number of responses differentiated the expansion from part (a) rather than the original expression; where the denominator was dealt with correctly, these responses could access at most two marks for this part.

Part (c)

Part (c) provided greater discrimination. Responses that rewrote the expansion from part (a) in the numerator typically identified the connection between the numerator and denominator and were able to proceed. Slips in handling the second term sometimes prevented the award of the accuracy mark, even though the result was not affected. Some responses attempted to use L'Hôpital's rule again, using the expansion from part (a) for the numerator, and so did not address the intended comparison between the two methods. Some responses did not attempt part (c), while others began by writing the expansion but stopped before including the denominator, so no further progress was made.

Question 8

This was another accessible question for most learners, with many responses gaining full marks. Where full marks were not achieved, the main reasons were slips in calculating the constant or errors in the final rearrangement to make W the subject.

Part (a)

Part (a) was tackled in a variety of ways – either working in terms of 'W' or in terms of 'x'. Most learners were able to form a correct linking equation using the chain rule, sometimes implied by substitution in one go using the equation (I). Commonly $\frac{dW}{dt}$ was found in terms of W and $\frac{dx}{dt}$, then substituted in and worked in terms of W with the final step being to change from W to x, but also popular was to find $\frac{dW}{dt}$ in terms of x and $\frac{dx}{dt}$, then substitute in and change all W terms correctly to x then rearrange. Another method seen was finding $\frac{dx}{dt}$ in terms of W and $\frac{dW}{dt}$ and then multiplying their differentiated W by the given equation for $\frac{dW}{dt}$.

This was a printed answer, and most learners showed sufficient working to show their method to achieve their correct answer. An error sometimes seen here was a problem with differentiating W or forming a wrong chain rule.

Part (b)

In part (b), the most common method was to find the integrating factor, multiply by it, integrate both sides, and then work towards the required answer. Separating the variables was much less common. A small number of responses used an auxiliary equation method, which also led successfully to the answer.

Once the integration was done, responses were split as to whether they changed from x to W before their next steps, though most work with x initially. Once they integrated both sides, most responses went on to apply the initial conditions to find the constant before rearranging to make W the subject, though there were a few responses who rearranged first and then found the constant. There were a variety of rearrangements for the final answer. Most proceeded to correctly rearrange the final equation to W as the subject from x, although a few made errors with the $\frac{3}{2}$ power, either taking powers of individual terms rather than the whole line (which is incorrect but was permitted the method for the attempt) or use of an incorrect index (which forfeited the method).

Decimal answers were rarely seen; responses generally gave exact answers. Some common errors for this question were incorrect integration of the exponential function, forgetting to multiply the RHS by the integrating factor, incorrect rearrangement when dividing by $e^{\frac{7t}{15}}$, not making W the subject and leaving as $W^{\frac{3}{2}}$ and numerical mistakes made in calculating the constant.

Question 9

This question provided effective discrimination at the end of the paper, while still allowing access in the earlier parts. Most responses were able to gain marks in the more routine parts (a) and (b). The latter stages of part (c), particularly beyond the initial method marks, provided greater discrimination, with fewer responses completing the question fully correctly.

Part (a)

Although part (a) was relatively routine, it did provide some challenge. Some responses did not first rearrange the equation into standard form. Although the correct eccentricity did proceed from the various a and b values used ($a^2 = \frac{1}{9}$ and $b^2 = \frac{1}{25}$ being common) the specification only covers equations in standard form and so these approaches were not credited. Learners should be reminded to work from the required standard form when identifying the parameters of a conic.

Some responses used $a^2 = \frac{1}{25}$ and $b^2 = \frac{1}{9}$ and had to force the result as it does not follow from these. Very few responses used an incorrect eccentricity formula; the more common reason marks were not awarded was the use of incorrect values for a and b.

Part (b)

Part (b) was the most successful part of the question overall. Most responses demonstrated the required method for finding the gradient at P and then used this to find the equation of the normal. The main issue was the omission of an intermediate step between substituting the values of *x* and *y* and giving the final equation, although responses using the $y = mx + c$ approach were less affected by this.

Most responses used implicit differentiation successfully. Responses using a parametric form were also often successful.

Part (c)

Part (c) provided greater challenge. There were many concise and fully completed solutions, but some responses made progress only in the early stages. Most responses were able to identify the focus, including by follow-through where incorrect values of a and b had been found in part (a). Most responses were also able to identify one suitable equation in a and b or other suitable combination of variables – usually the $b^2 = 16 - a^2$ equation (or follow through with their e). The second equation, obtained by substituting the coordinates of P into the hyperbola equation, was less frequently identified. Some responses reached this equation through a longer route, such as solving the hyperbola and ellipse equations simultaneously before substituting the relevant coordinate after eliminating either x or y. Various approaches using parametrisations of the hyperbola were also seen; some were completed successfully, although these methods were more prone to error.

Responses that identified two suitable equations usually went on to eliminate one variable and solve. Where a suitable single-variable equation was obtained, calculator methods were commonly used successfully. In fully correct solutions, this led to the correct values of a and b. In other responses, errors in rearranging or in the values of a and b led to incorrect final values. A small number of responses stopped after finding the squares of a and b, so only the final mark was not awarded.

Part (d)

Where responses had not formed an equation in part (c), only a small number went on to attempt part (d). For responses that did continue, the first mark was often awarded, as the method for finding the gradient was usually demonstrated clearly. The final two marks depended on having the correct values from part (c). Some responses gave the correct gradient after using incorrect values from part (c), but as the gradient did not follow from their working, only the first mark could be awarded. Some responses reached an expression for the derivative in terms of x and y but did not substitute the coordinates, so the first mark was not secured.

For the second mark, many responses accessed this by finding the equation of the line accurately. Stronger responses recognised that equal gradients, together with the shared point P, were sufficient to show that the lines were the same. Where marks were not awarded, this was often because the necessary conclusion was not stated clearly. Some responses gave the equation of the line but did not include a supporting statement explaining that the two lines were parallel and passed through the same point, or that the equations were equivalent. Learners should be reminded to include clear deductions when a result depends on a geometrical relationship rather than calculation alone.